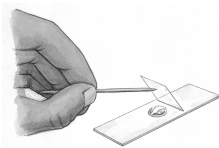


Preparing a stained wet mount

Answers

Jumbled steps

1. The following diagrams show the jumbled steps used to prepare a wet mount of onion epidermis. Place the code letters in the correct sequence.



A. Lower a coverslip over the onion skin and water droplet.



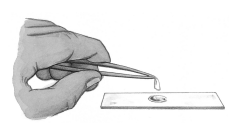
B. Peel a section of onion epidermis.



C. Soak up excess water with a tissue.



D. Add a drop of water to a microscope slide.



E. Place onion skin in drop of water on the slide.

Correct sequence: B, D, E, A, C

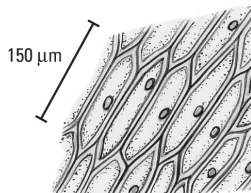
Staining cells

2. It is much easier to see the internal structure of cells by staining the tissue. The following materials are available to prepare a stained wet mount of onion epidermis.
dropping bottle of methylene blue or other similar stain, microscope, microscope slide, coverslip, small section of a peeled onion, forceps, toothpick, dropping bottle of water

- (a) Write a method for your investigation.

1. Peel a section of onion epidermis and remove a section of epidermis with forceps.....
2. Add a drop of methylene blue to a slide.....
3. Place the onion epidermis in the stain.....
4. Cover with a coverslip using a toothpick to lower the coverslip.....
5. Remove excess stain with a tissue.....

- (b) The drawing below left was made of the stained onion epidermal tissue.



- (i) The nucleus and cell walls take up the methylene blue stain. Use a blue pencil to colour these components in the drawing.
(ii) Use the scale to estimate the greatest width of a cell.

- (iii) The experiment was repeated with iodine stain, which strongly stains starch grains a deep blue-black colour. Use coloured pencils to colour the starch grains, which are scattered in the cytoplasm in the cells at the right.



- (i) Students colour the cell walls and nuclei blue..... (ii) 35–40 μm
(iii) Many tiny starch grains should be coloured blue-black.....